UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

PHYSICS **5054/01**

Paper 1 Multiple Choice **May/June 2007**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

0188762188

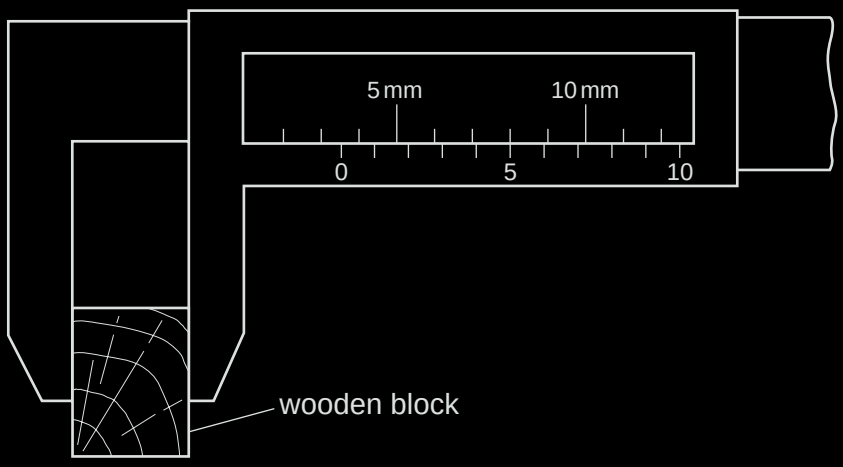
READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.



What is the width of the block?

A 3.5 mm **B** 5.3 mm **C** 8.0 mm **D** 8.5 mm

2 Which of the following correctly lists one scalar and one vector quantity?

	scalar quantity	vector quantity
A	displacement	work
B	energy	force
C	force	acceleration
D	velocity	mass

3 A student drops a table-tennis ball in air.

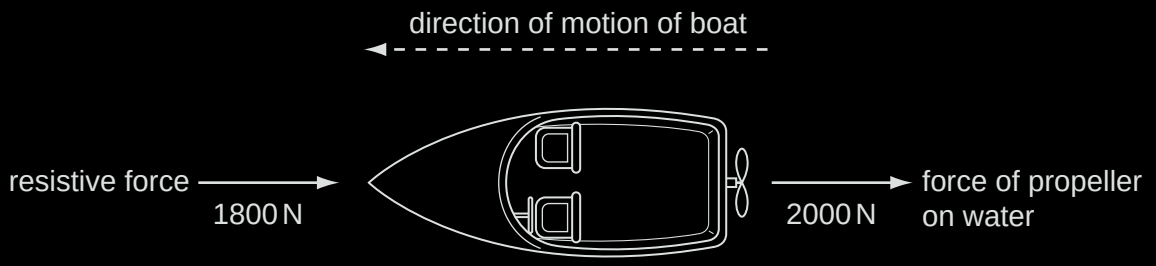
What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

4 A car is brought to rest in 5 s from a speed of 10 m / s.

What is the average deceleration of the car?

A 0.5 m / s^2 **B** 2 m / s^2 **C** 15 m / s^2 **D** 50 m / s^2



According to Newton's third law, what is the forward force on the propeller due to the water?

A 3800 N **B** 2000 N **C** 1800 N **D** 200 N

6 A student does an experiment to estimate the density of an irregularly-shaped stone.

Which items of equipment are needed?

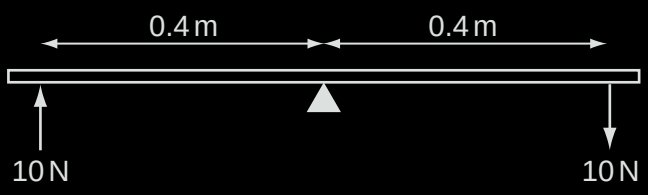
A a balance and a measuring cylinder containing water

B a balance and a ruler

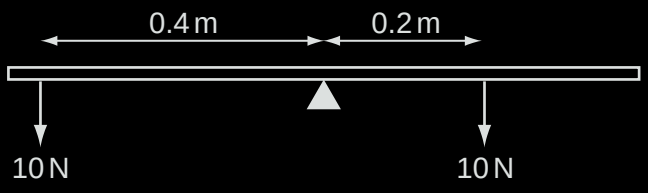
C a ruler and a measuring cylinder containing water

D only a measuring cylinder containing water

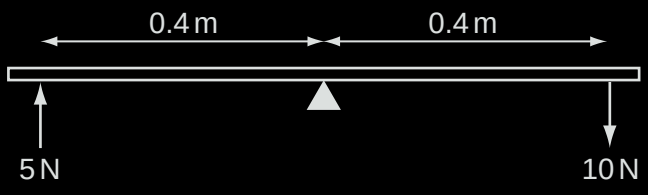
A



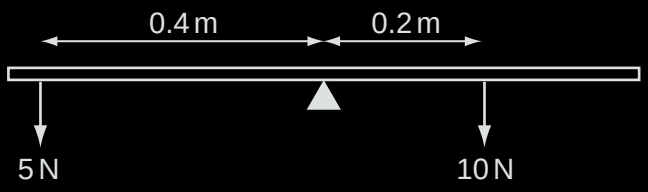
B

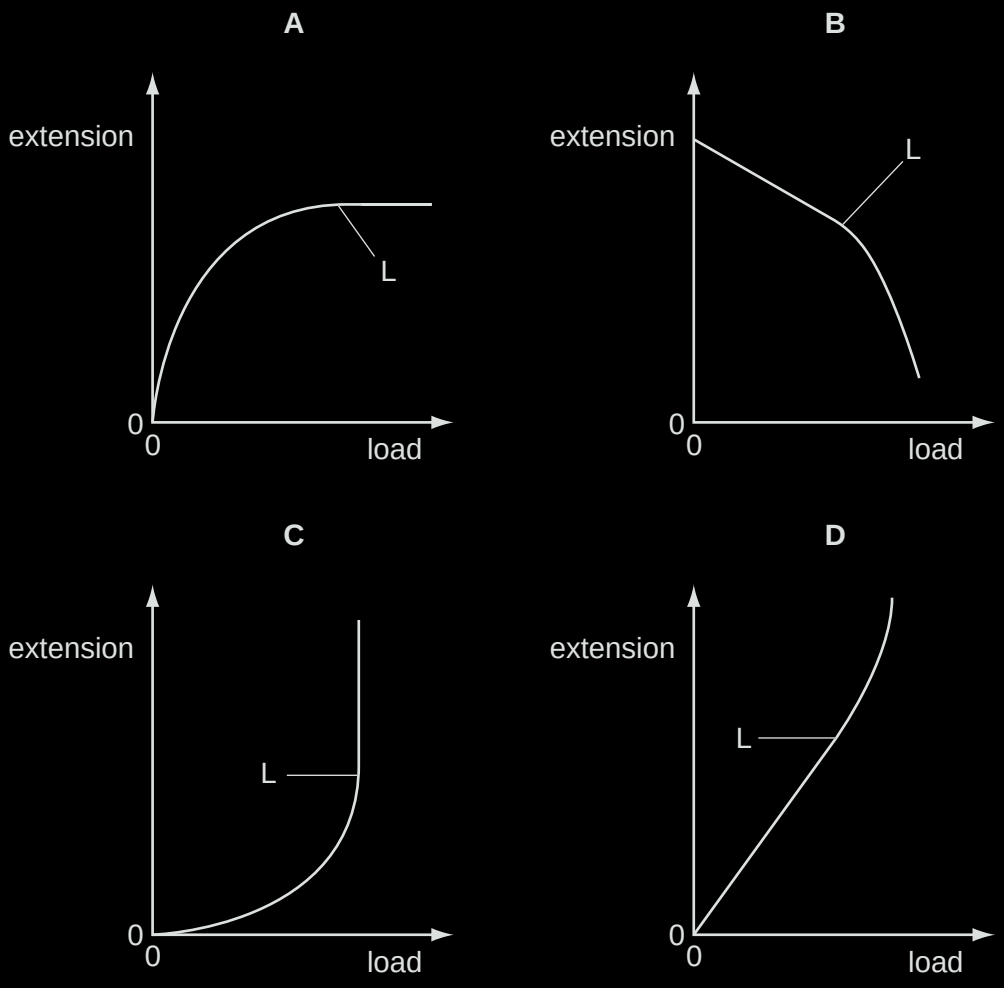


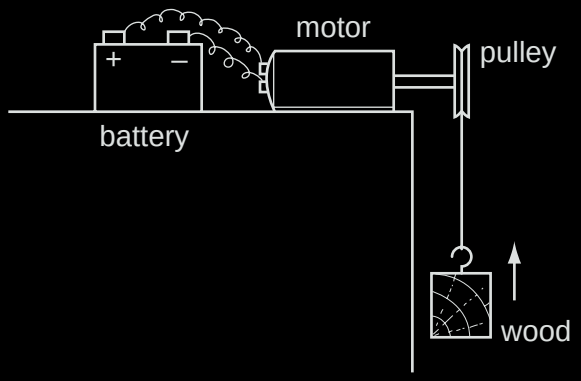
C



D




A
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What is the overall energy change taking place?

A chemical → gravitational potential

B gravitational potential → electrical

C gravitational potential → kinetic

D kinetic → gravitational potential

10 Four students exercise in a gym.

Which student does the most work?

	exercise time / s	power developed / W
A	50	250
B	100	150
C	200	200
D	250	30

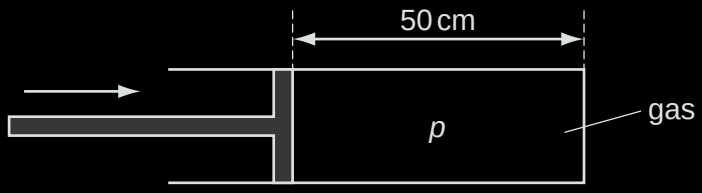
11 Which formula gives the efficiency of an energy conversion?

A efficiency = total energy input - useful energy output

B efficiency = useful energy output × total energy input

C $\text{efficiency} = \dfrac{\text{useful energy output}}{\text{total energy input}}$

D $\text{efficiency} = \dfrac{\text{total energy input}}{\text{useful energy output}}$



The piston is pushed in a distance of 30 cm, so that the length of the gas column is now 20 cm. The temperature of the gas does not change.

What is the new pressure of the gas?

A 0.4 *p* **B** 0.6 *p* **C** 1.5 *p* **D** 2.5 *p*

13 A liquid is heated.

Which statement is **incorrect**?

A The molecules expand.

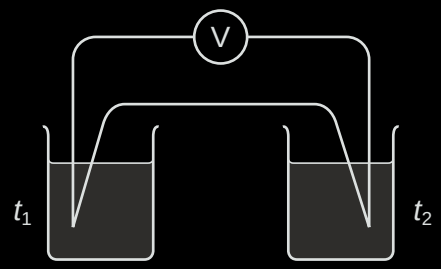
B The molecules gain energy.

C The molecules move further apart.

D The molecules move faster.

14 What describes the molecular structure of a liquid?

	distance between the molecules	motion of the molecules	strength of forces between the molecules
A	close together	stationary	very strong
B	close together	random	fairly strong
C	far apart	stationary	fairly strong
D	far apart	random	weak



What are the values of t_1 and t_2 when the thermometer is calibrated at the steam point?

	t_1	t_2
A	0 °C	0 °C
B	0 °C	100 °C
C	25 °C	0 °C
D	25 °C	125 °C

16 What is the definition of *heat capacity*?

A the quantity of heat required to raise the temperature of an object through 1 °C

B the quantity of heat required to raise the temperature of 1 kg of a substance through 1 °C

C the quantity of heat required to convert an object from solid to liquid without a change in temperature

D the quantity of heat required to change 1 kg of a substance from solid to liquid without a change in temperature

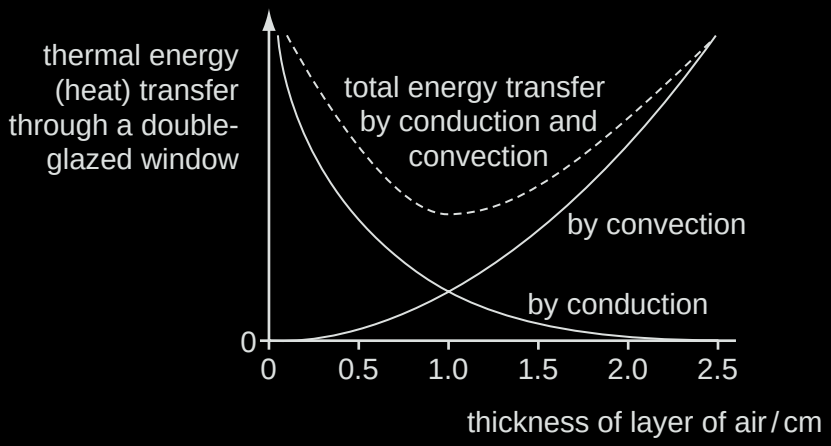
17 Fillings in teeth should be made from a material which

A expands more than the tooth when heated.

B expands by the same amount as the tooth when heated.

C expands less than the tooth when heated.

D does not expand when heated.



Which thickness of air produces the smallest energy transfer, and why?

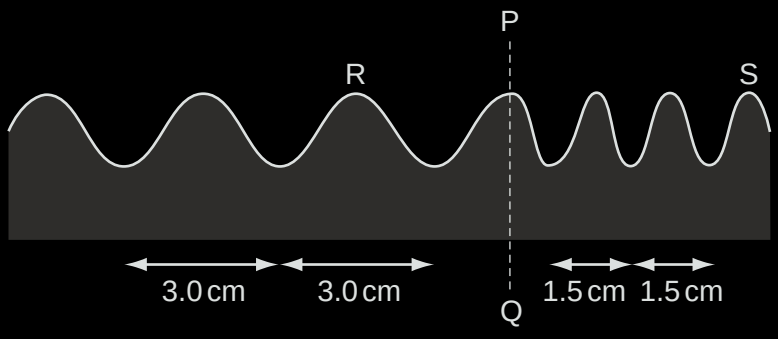
A 0.5 cm because there is little convection

B 1.0 cm because the total thermal energy transfer is least

C 1.5 cm because the total thermal energy transfer is small and conduction is low

D 2.0 cm because there is little conduction

19 The diagram shows a water wave in a ripple tank.

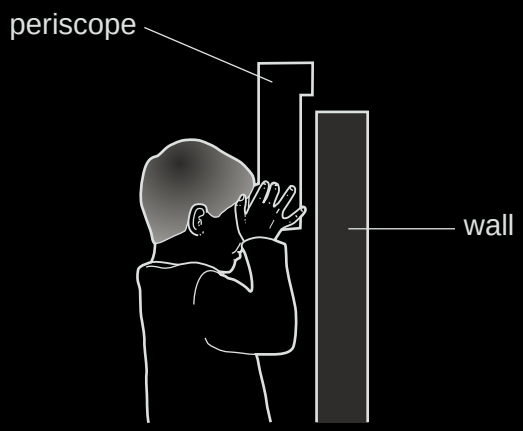


The wave has a speed of 12 cm / s at R.

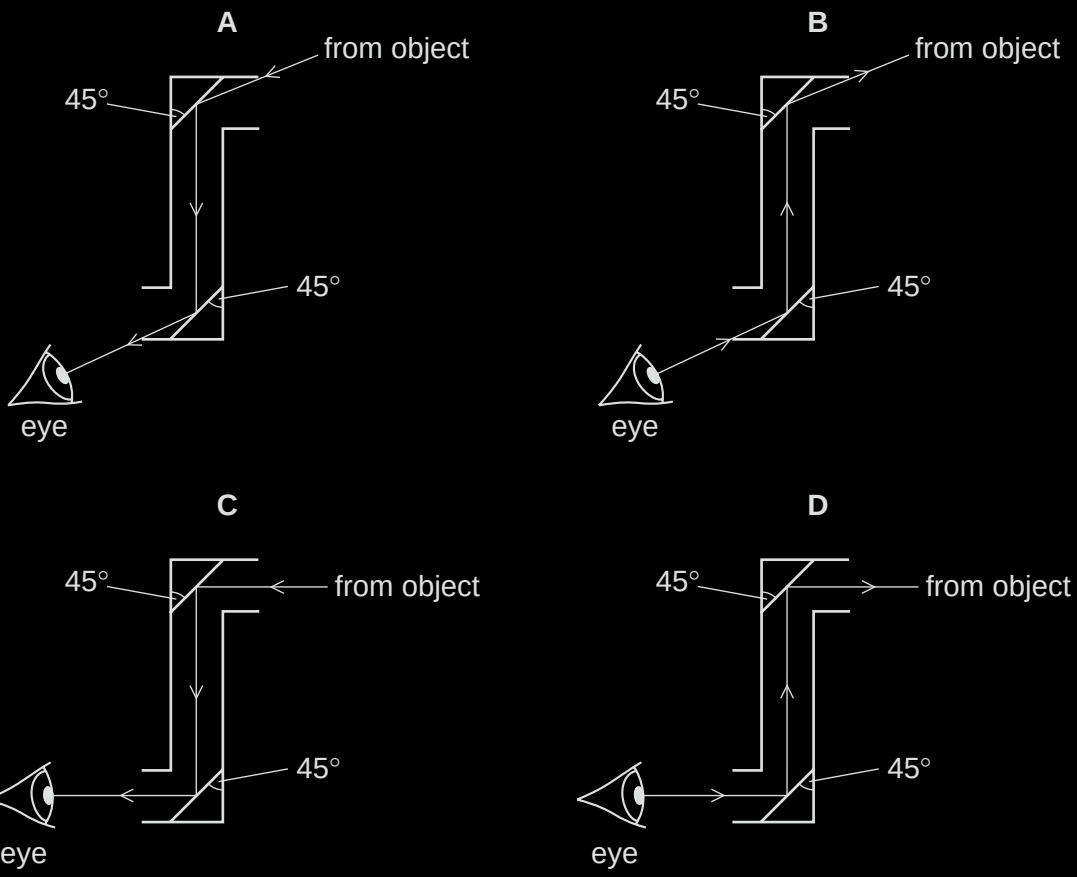
The wave crosses a boundary PQ where the distance between crests changes from 3.0 cm to 1.5 cm.

What is the speed of the wave at S?

A 3.0 cm / s **B** 6.0 cm / s **C** 12 cm / s **D** 24 cm / s



Which diagram shows a correctly drawn ray of light from the object?



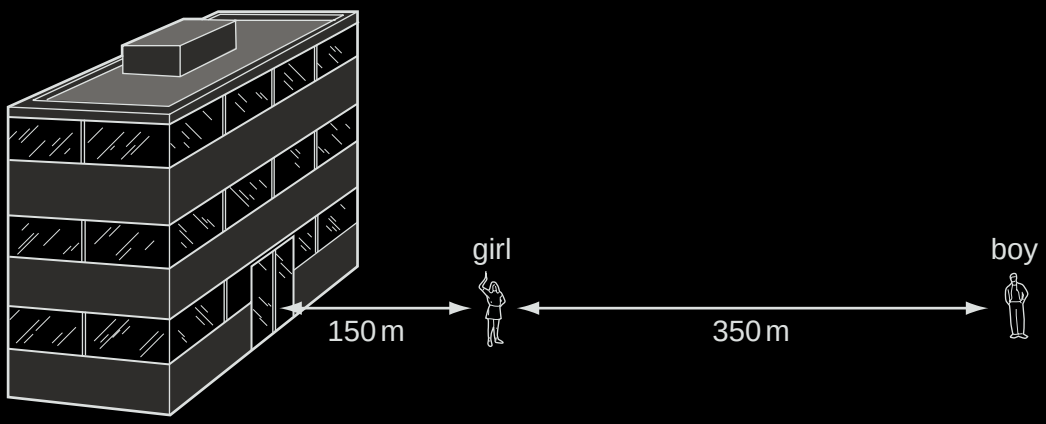
21 What happens to light as it passes from glass into air?

A Its frequency decreases because its speed decreases.

B Its frequency increases because its speed increases.

C Its wavelength decreases because its speed decreases.

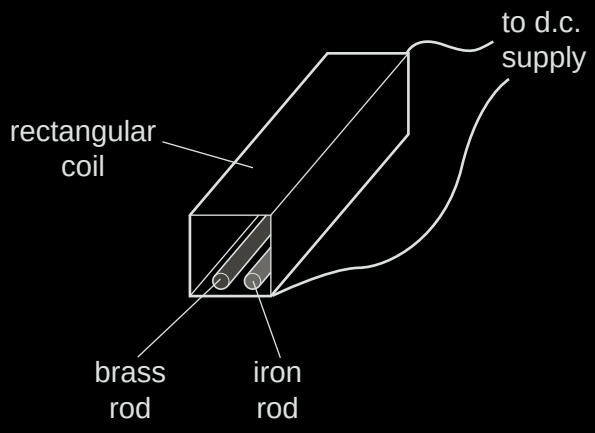
D Its wavelength increases because its speed increases.



From this information, what is the speed of sound in air?

A 300 m / s **B** 350 m / s **C** 500 m / s **D** 650 m / s

24 The diagram shows a brass rod and an iron rod beside each other at the bottom of a rectangular coil.



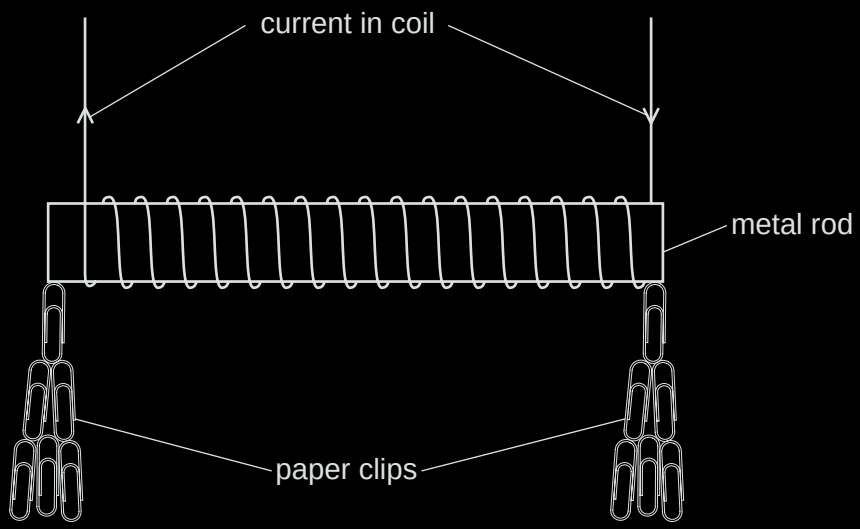
What happens when a d.c. current passes through the coil?

A Only the brass rod is magnetised.

B Only the iron rod is magnetised.

C The two rods attract each other.

D The two rods repel each other.

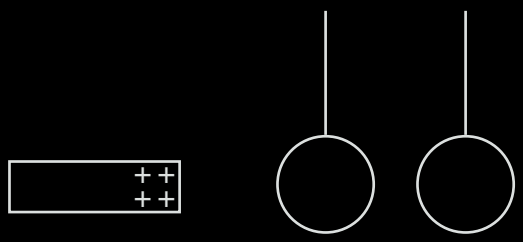


Each rod is used to pick up as many paper clips as possible. The current is then switched off.

The table gives the results of the experiment.

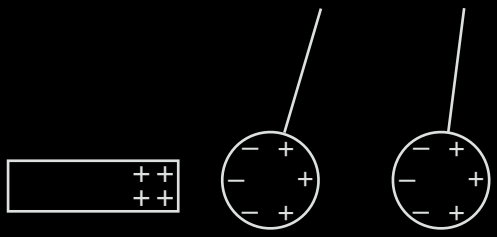
Which rod is the most suitable core for a coil in a circuit breaker?

	number of paper clips picked up when there is a current in the coil	number of paper clips still attached after the current is switched off
A	1	0
B	20	2
C	35	0
D	35	30

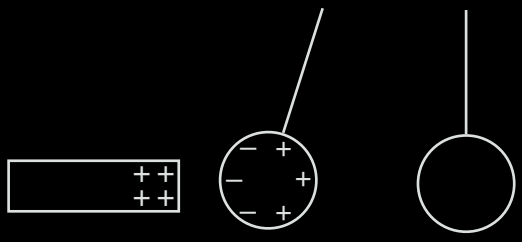


Which diagram shows what happens to the spheres?

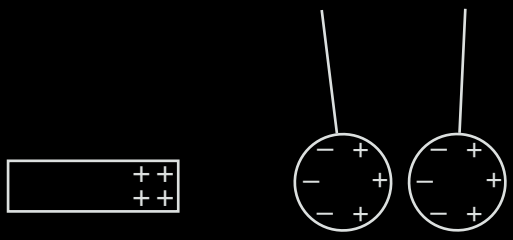
A



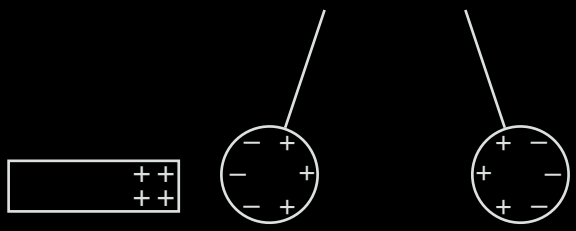
B



C

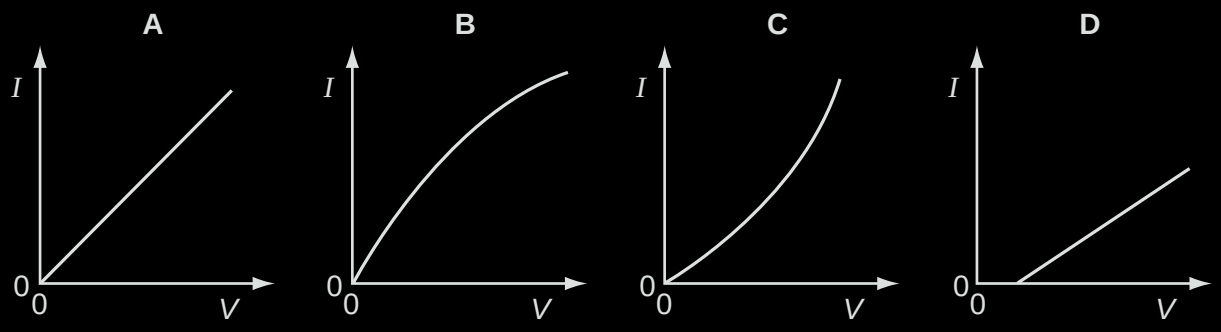


D

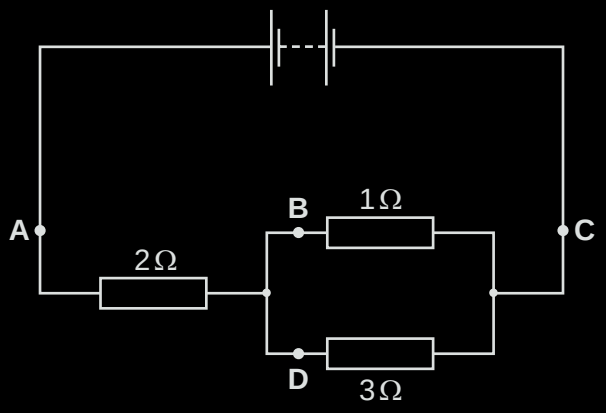


27 Which factors will both increase the resistance of a wire in a circuit?

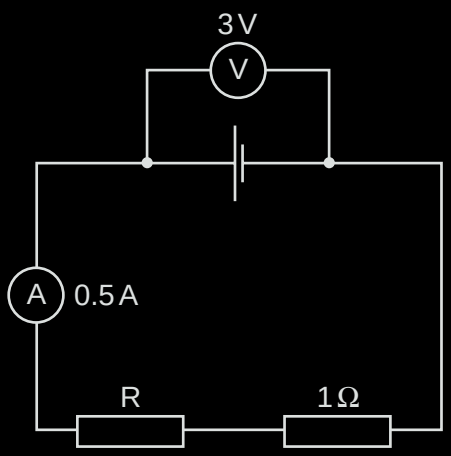
	size of wire	temperature of wire
A	longer	lower
B	shorter	lower
C	thicker	higher
D	thinner	higher



29 At which point in the circuit is the current the smallest?



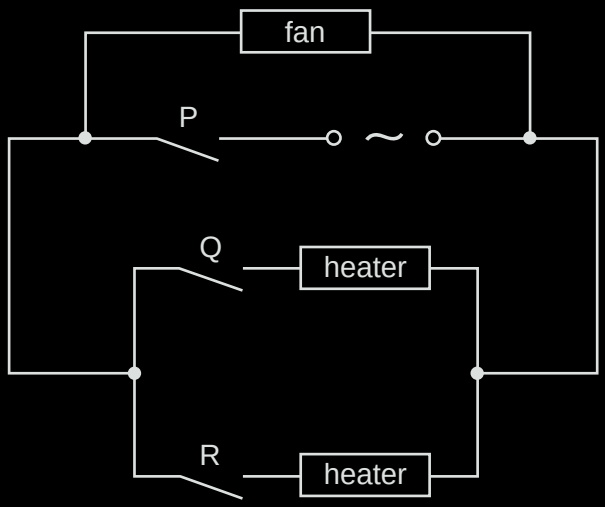
30 The diagram shows a circuit.



The ammeter has negligible resistance.

What is the resistance of the resistor R?

A 0.5Ω **B** 1.5Ω **C** 5Ω **D** 6Ω

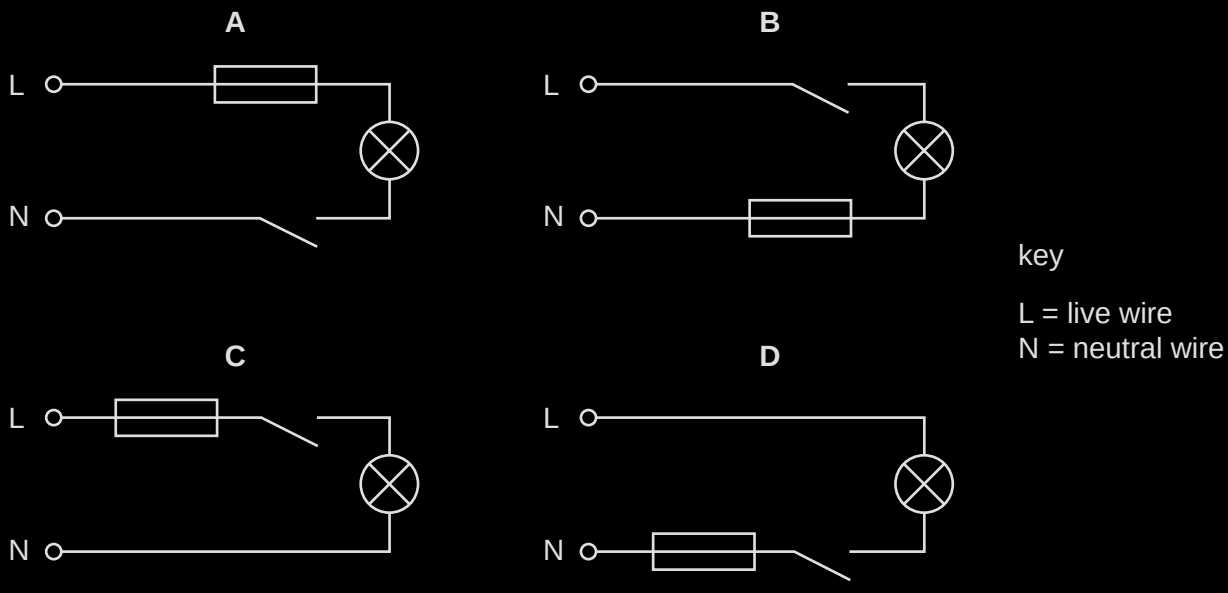


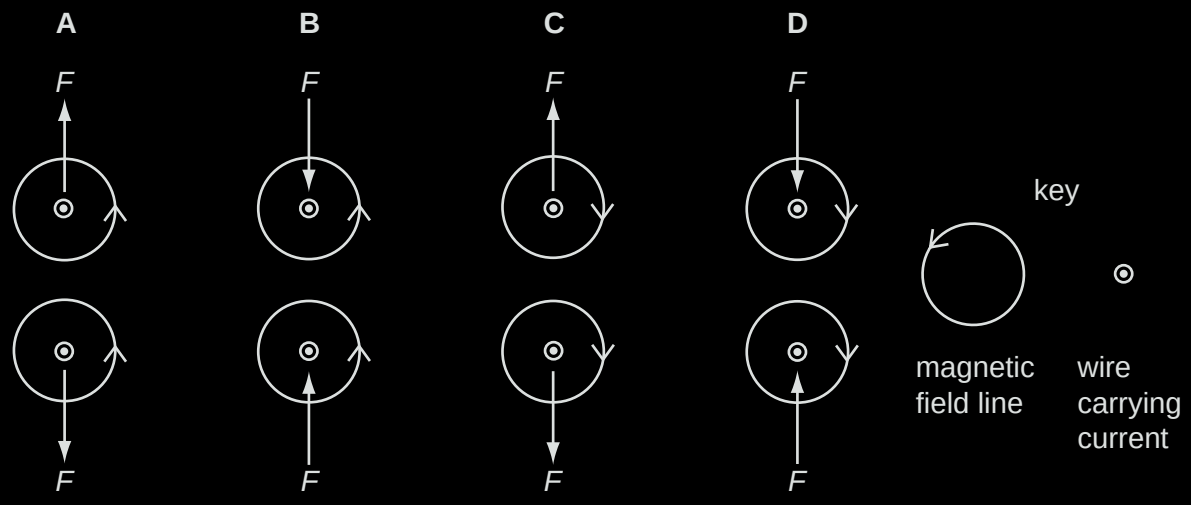
The fan has a power rating of 0.1 kW and the heaters each have a rating of 0.4 kW. The cost of electricity is 8 cents/kW h.

What is the cost of running the dryer for two hours with switches P and Q closed and switch R open?

A 1.6 cents **B** 3.2 cents **C** 6.4 cents **D** 8.0 cents

32 Which circuit shows the correct positions for the fuse and the switch in the lighting circuit of a house?





34 In a d.c. motor, the coil is wound on a soft iron cylinder.

Why is soft iron used?

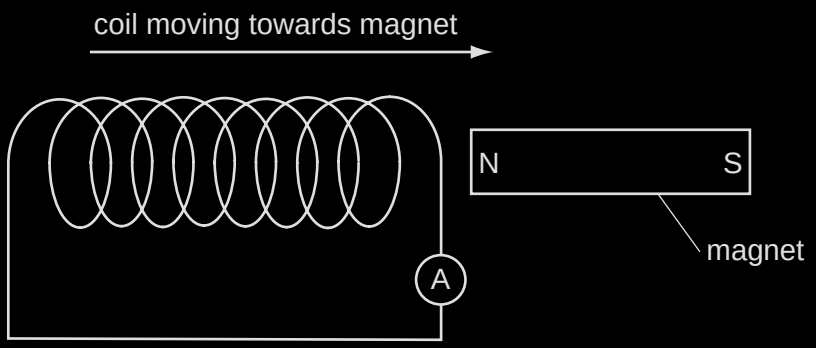
A to increase the strength of the magnetic field

B to increase the weight of the coil

C to insulate the coil from the magnet

D to prevent the coil from spinning too quickly

35 The diagram shows how a magnet and a coil may be used to induce an electric current.



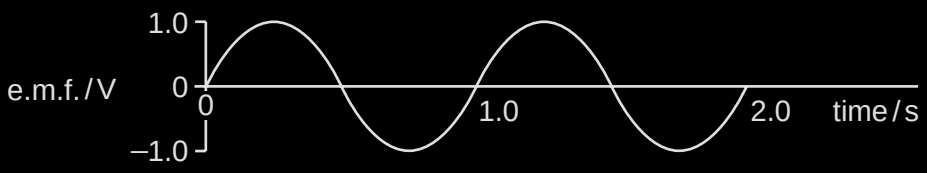
How could the ammeter reading be increased?

A Move the coil more slowly.

B Put a resistor in series with the ammeter.

C Turn the magnet round, then move the coil.

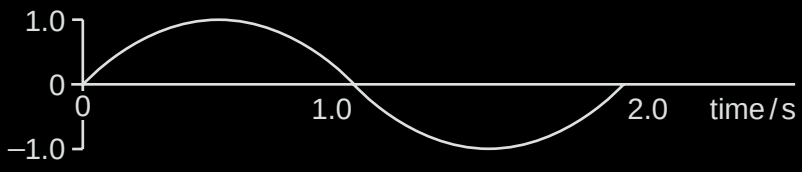
D Use a coil with more turns.



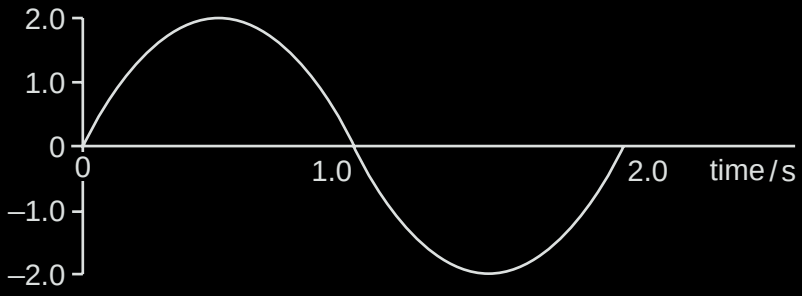
The speed of the generator is doubled.

Which graph best represents the new output?

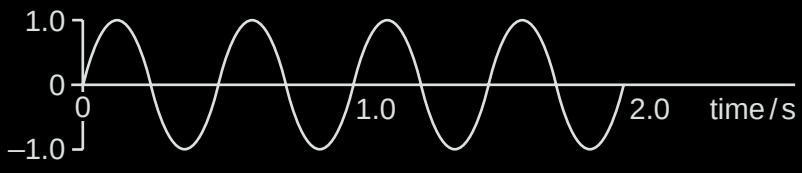
A



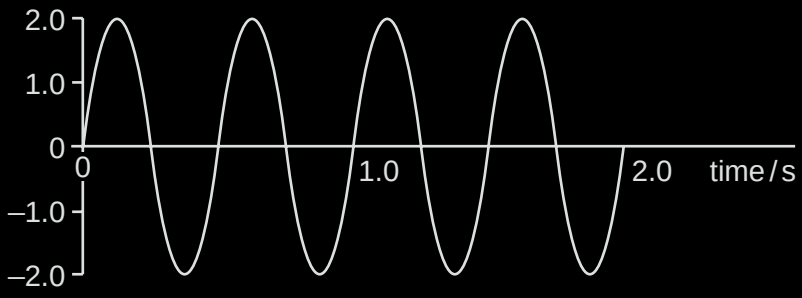
B

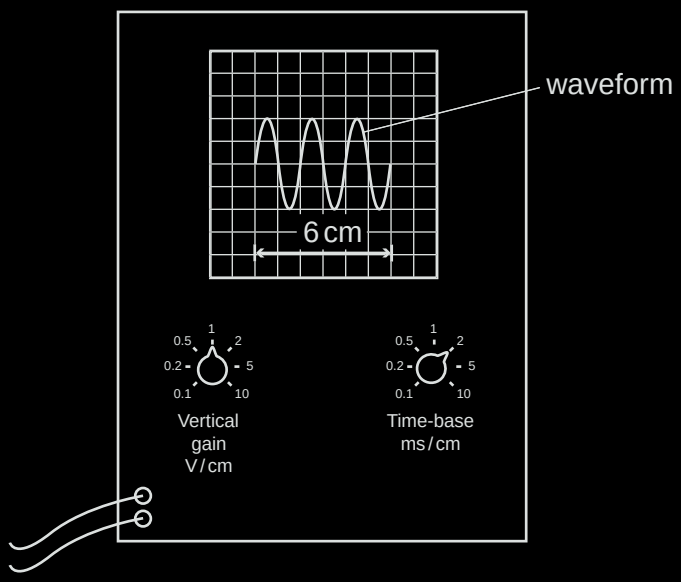


C



D





What is the time taken for one cycle of the waveform?

A 1 ms **B** 2 ms **C** 4 ms **D** 6 ms

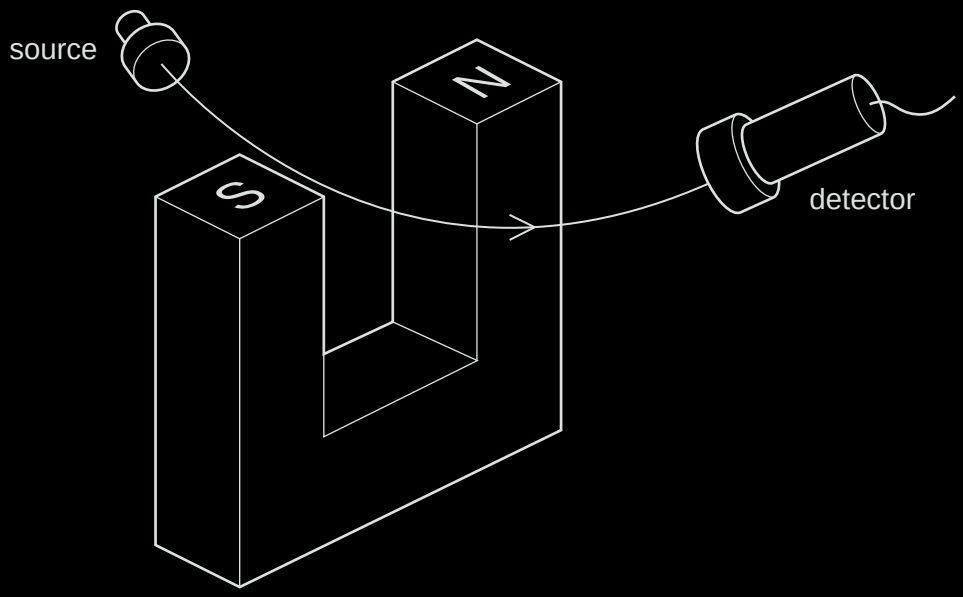
38 Which statement is true for all three types of radioactive emission (alpha-particles, beta-particles and gamma-rays)?

A They are completely absorbed by a thin aluminium sheet.

B They are deflected by electric fields.

C They emit light.

D They ionise gases.



Which type of radiation is being detected?

A alpha-particles only

B beta-particles only

C gamma-rays only

D alpha-particles and beta-particles

40 A nuclide has the notation $^{48}_{23}X$.

Which line in the table describes a different isotope of this nuclide?

	proton number (atomic number)	nucleon number (mass number)
A	23	50
B	24	48
C	48	24
D	50	23